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$^{40}Ar/^{39}Ar$ ages and paleomagnetism of São Miguel lavas, Azores

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Abstract

We present new $^{40}Ar/^{39}Ar$ ages and paleomagnetic data for São Miguel island, Azores. Paleomagnetic samples were obtained for 34 flows and one dike; successful mean paleomagnetic directions were obtained for 28 of these 35 sites. $^{40}Ar/^{39}Ar$ age determinations on 12 flows from the Nordeste complex were attempted successfully: ages obtained are between 0.78 Ma and 0.88 Ma, in contrast to published K–Ar ages of 1 Ma to 4 Ma. Our radiometric ages are consistent with the reverse polarity paleomagnetic field directions, and indicate that the entire exposed part of the Nordeste complex is of a late Matuyama age. The duration of volcanism across São Miguel is significantly less than previously believed, which has important implications for regional melt generation processes, and temporal sampling of the geomagnetic field. Observed stable isotope and trace element trends across the island can be explained, at least in part, by communication between different magma source regions at depth. The $^{40}Ar/^{39}Ar$ ages indicate that our normal polarity paleomagnetic data sample at least 0.1 Myr (0–0.1 Ma) and up to 0.78 Myr (0–0.78 Ma) of paleosecular variation and our reverse polarity data sample approximately 0.1 Myr (0.78–0.88 Ma) of paleosecular variation. Our results demonstrate that precise radiometric dating of numerous flows sampled is essential to accurate inferences of long-term geomagnetic field behavior. Negative inclination anomalies are observed for both the normal and reverse polarity time-averaged field. Within the data uncertainties, normal and reverse polarity field directions are antipodal, but the reverse polarity field shows a significant deviation from a geocentric axial dipole direction. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; magnetic field; secular variations; Ar-40/Ar-39; absolute age; Azores

1. Introduction

Over the past three decades many studies of paleosecular variation from lavas (PSVL) around the world have contributed to the compilation of a comprehensive global data base [1] and also to more restricted global data sets [2,3]. Given a global distribution of PSVL data, we can in theory, model both the time-averaged structure of the geomagnetic field (TAF) and the statistics of the temporal variations about this mean field structure (paleosecular varia-

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tion, PSV). However, global field modeling efforts to date have been hampered by data quality and temporal sampling issues associated with available PSVL data. Many existing paleomagnetic data were collected and processed more than 20 years ago. Consequently, modern laboratory procedures were not applied, and reported paleomagnetic directions are often contaminated by unremoved overprints. In addition to data quality issues, temporal sampling of the existing 0–5 Ma global lava flow data set is very uneven. To obtain estimates of PSV and the TAF, lava flows sampled from any given location must span a sufficiently long time interval to accurately describe the statistical behavior of the field. This is a complicated issue as there is no definitive time interval that one can say is required to get an 'average' state for the field. Regardless of whether such a time interval can be sensibly defined and agreed upon, the time period spanned by lava flow sequences in PSVL studies varies from one location to another. The temporal distribution of flows is often poorly known for an individual investigation, being based primarily on stratigraphic relations, previous radiometric dating of flows in the sequence which may or may not be sampled in the PSVL study, and occasionally radiometric dating of a few flows sampled in the PSVL study. Thus inferences of the time period sampled in a given PSVL investigation may be grossly in error.

The goals of our study were to address some of these data quality and temporal sampling issues via paleomagnetic sampling and radiometric dating of lava flows from São Miguel island in the Azores. The abundance of Pleistocene and Pliocene volcanism makes São Miguel particularly suited to paleomagnetic sampling aimed at studying the TAF and PSV. Paleomagnetic data from the Azores are already available [4–6]; however, data from only 20 Terceira lava flows [7] are suitable for use in global geomagnetic field modeling [3,8], and radiometric age determinations for the flows themselves were not performed. Our objectives were to characterize the TAF and PSV for the last 4 Ma, by sampling previously dated flows, flows of the Nordeste complex and by obtaining additional radiometric dates. Here we present our paleomagnetic data and $^{40}Ar/^{39}Ar$ results and discuss their implications for the timing and duration of volcanism on São Miguel.

2. São Miguel Island, Azores

Nine islands form the Azores archipelago, which lies in the vicinity of the N. America–Europe–Africa triple junction. Two of the islands lie west of the Mid-Atlantic ridge (MAR), whilst the remaining seven islands (including São Miguel) lie east of the ridge. To the west the MAR is an active spreading center and to the east the Gloria Fault is a right-lateral strike slip fault. The Africa–Europe plate boundary in this region is probably coincident with the Terceira rift, a slow spreading center underlying the Azores. The tectonics and the geochemistry of the Azores region are complex and have been interpreted as broadly consistent with a hotspot origin for the Azores platform [9]. The island of São Miguel is particularly unusual geochemically, displaying strong intra-island heterogeneities. Stable isotope and trace element data show a gradual west–east trend across the island; they are consistent with a MORB-like depleted mantle source for flows in the west (presumably related to the MAR), and an enriched mantle source and higher degrees of partial melting for flows in the east [10,11].

Extensive geological mapping of São Miguel [12–15], along with radiometric dating [16–18] has led to the identification of six volcanic zones ([14]; Fig. 1), associated primarily with the four named major volcanoes on the island. The Nordeste shield of eastern São Miguel comprises the oldest exposed units on the island [13], and has been subdivided into five stratigraphic sequences: lower basalts, Nordeste ankaramites, upper basalts, tristanites and trachytes. Existing K–Ar age determinations for the complex are 4.01 Ma and 3.17 Ma for two lower basalt flows, 1.86 Ma for the upper basalts and 1.28 Ma, 1.23 Ma, and 0.95 Ma for the tristanites and trachybasalts [16]. Three stratovolcanoes (Furnas, Agua de Pau and Sete Cidades) are Pleistocene in age and have been mapped in detail by Moore [14]. Historical eruptions have been recorded for all three volcanoes. Ankaramites, alkali olivine basalts, hawaiites, mugearites, tristanites, along with trachyte domes, flows and pyroclastic deposits are exposed. Radiocarbon age determinations ranging from 0 to 40,000 years for charcoal found in paleosols below 63 lava flows and pyroclastic deposits [18], and seven K–Ar age determinations spanning 46,000–210,000 years

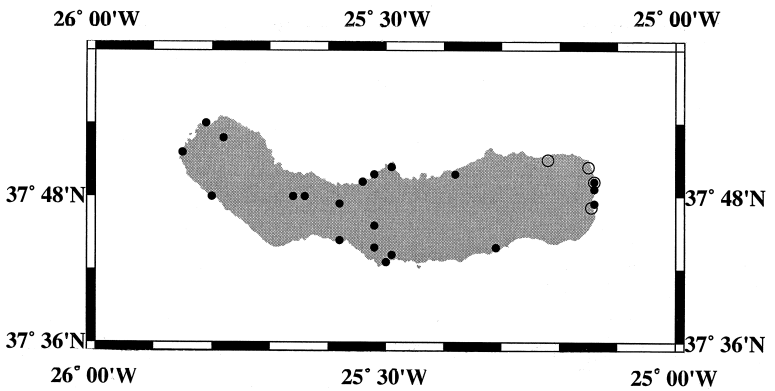


Fig. 1. Map of São Miguel with volcanic provinces (after Moore [15]). Sampling sites numbered; dots are normal polarity sites, circles reverse polarity sections. Six major volcanic zones (four named) delineated by dashed lines.

[17] provide important constraints on the ages of various volcanic units.

3. Paleomagnetic sampling

Cores were drilled and oriented in situ with a magnetic compass, sampling 34 flows and one dike (Fig. 1). Where possible, cores were also oriented with a sun compass (approximately 53% of the cores), with at least one sun compass measurement made at each site. A minimum of 10 cores per flow were drilled, most of these yielded 2–3 specimens per core. At each site, local stratigraphic relations were documented.

Age constraints on our nineteen Brunhes polarity flows are good as reliable radiometric dates are available for the majority of our Brunhes sites (Table 1; [17,18]). Other known Brunhes-age flows sampled were close to dated sites and were selected on the basis of mapped units [15]. We initially sampled nine flows (AZ01–AZ11, samples for geochemical analyses only were collected at AZ05 and AZ06) spanning the last 200,000 year. Site AZ12 is believed to be early Brunhes in age, on the basis of geological mapping [15] and K–Ar radiometric dates of 0.76 Ma and 0.78 Ma for two nearby flows at Maia (Forjaz, pers. commun.; Geothermal Project data). We later sampled nine additional Brunhes age flows, all in the age range 0–10 ka, to enable us to compare PSV and the TAF over different time intervals during the Brunhes period. The availability of both radiometric dates and the extensive previous geological mapping was critical in guiding our sampling of Brunhes flows. São Miguel is heavily vegetated and outcrops in western São Miguel are generally isolated. Table 1 lists our sites in order of increasing age, on the basis of geological mapping [15], radiometric age determinations and our own observations of local stratigraphy.

Sites AZ13–AZ28 were chosen in order to sample older flows (on the basis of mapping; Moore [15]) from four Nordeste cliff sections (eastern São Miguel) —Praia Lombo da Gorda (AZ13–AZ19), Porto de Nordeste (AZ21–AZ22) and nearby site AZ20, Ribeira Despe-te que Suas (AZ23–AZ26), and Ribeira do Guilherme (AZ27–AZ28). Most large cliff sections of the Nordeste complex appeared to be composed primarily of reverse polarity flows, often cut by normal polarity hawaiite dike swarms of unknown age, such as at site AZ15. The well-exposed cliff sections enabled documentation of stratigraphic relations between sites within each of the four sections. In each section sampled the lowest sites stratigraphically were AZ13, AZ21, AZ23 and AZ28, respectively. Based on prior geological mapping and our own observations, site AZ21 appears to be lower than AZ13; though the time interval during which the entire Porto de Nordeste section was emplaced probably overlaps with the time interval for emplacement of the Praia Lombo da Gorda section. The base of the Porto de Nordeste section was previously dated at 4 Ma [16]. The Praia Lombo da Gorda and Porto de Nordeste sections comprise mainly the lower basalts of [13]; the ankaramite flows at sites AZ20 and AZ22 are inferred to be the younger Nordeste ankaramites.

Table 1

Site	Location (37°N; 25°W) [a]	Unit [b]	Age (ka) [c]	N_c	N_m	N	D	I	k	α_{95}
Known Brunhes age sites										
AZ38	44.46′; 34.95′	f9b	0.345	12	9	5	15.0	48.4	728	2.8
AZ30	49.15′; 32.55′	f9b	0.434	11	6	5	0.7	58.2	246	4.9
AZ01	48.26′; 48.21′	f8b	0.91	12	5	(5)				
AZ29	47.53′; 34.81′	f8b	1.01 ± 0.12	11	5	5	5.6	35.0	271	4.7
AZ31	49.22′; 32.34′	f7h	1.79 ± 0.15	11	5	5	352.8	57.3	253	4.8
AZ37	48.25′; 39.45′	f7b	2.46 ± 0.22	10	9	9	359.9	49.2	1046	1.6
AZ36	48.24′; 38.49′	f7b	2.80 ± 0.12	10	7	7	10.0	54.9	347	3.2
AZ32	49.65′; 31.26′	f6y	4.99 ± 0.10	12	7	6	352.2	56.0	279	4.0
AZ34	43.98′; 31.20′	f5a	6.52 ± 0.10	10	8	7	0.1	57.6	233	4.0
AZ33	45.83′; 32.16′	f5a	8.70 ± 0.20	12	8	(4)				
AZ02	48.21′; 48.15′	f4m	21	11	6	(4)				
AZ03	53.89′; 48.85′	f4a	10–30	11	6	(5)				
AZ04	51.51′; 51.16′	f3r	74 ± 6	12	6	(6)				
AZ09	42.95′; 29.20′	f3t	30–100	12	11	11	6.5	17.4	874	1.5
AZ10	43.83′; 18.74′	f3b	93 ± 9	11	6	6	337.6	39.2	295	3.9
AZ11	50.15′; 29.12′	o2t	103	11	10	6	9.7	59.6	511	3.0
AZ08	42.78′; 29.75′	d2t	181	11	5	(4)				
AZ07	52.80′; 46.20′	f1t	210 ± 8	13	7	(7)				
AZ12	49.98′; 22.96′	f1m		14	10	10	356.4	59.1	199	3.4
Nordeste sites										
AZ28	50.53′; 09.06′	f1k		11	11	10	180.0	−38.9	475	2.2
AZ27	50.53′; 09.06′	f1k	[776 ± 12]*	11	11	11	166.7	−45.2	700	1.7
AZ26	50.96′; 13.38′	f1f	[793 ± 08]	11	10	5	170.6	−35.0	107	7.4
AZ25	50.92′; 13.38′	f1f	[814 ± 17]	11	10	9	184.2	−42.2	166	4.0
AZ24	50.92′; 13.38′	f1f	[817 ± 14]	11	10	9	189.6	−42.9	817	1.8
AZ23	50.92′; 13.38′	f1f	[785 ± 11]*	12	11	10	167.0	−47.5	942	1.6
AZ20	48.65′; 08.63′	f1f	see legend	10	10	10	350.3	48.3	258	3.0
AZ22	49.42′; 08.14′	f1f		11	9	9	329.2	71.1	472	2.4
AZ15	47.14′; 08.60′	f1f		13	10	9	8.8	67.8	584	2.1
AZ19	47.10′; 09.10′	f1b	[852 ± 10]	12	12	12	162.2	−62.1	747	1.6
AZ17	47.18′; 08.74′	f1f	[819 ± 17]*	13	10	9	172.0	−46.8	209	3.6
AZ16	47.17′; 08.86′	f1f	[843 ± 09]	12	11	11	191.4	−38.0	87	4.9
AZ18	47.06′; 08.74′	f1f	[849 ± 14]	12	11	11	182.7	−42.8	350	2.4
AZ14	47.14′; 08.58′	f1f	[846 ± 11]	12	11	11	139.1	−22.7	337	2.5
AZ13	47.23′; 08.57′	f1f	[835 ± 11]*	11	10	8	199.5	−42.5	676	2.1
AZ21	48.20′; 08.12′	f1f	[878 ± 45]	12	10	10	180.7	−47.0	555	2.1
All normal data			Non-fisherian			15	359.1	52.6	29	7.2
All reverse data			Non-fisherian			13	175.8	−43.7	30	7.6

[a] Min relative to (37°N, 25°W). From GPS, except AZ07.

[b] From map of Moore [15].

[c] Previous radiometric dates in ka [17,18]. Where available, uncertainties in the ages are given. Previous age determination for site AZ20 was 1.86 Ma [16]. Numbers in brackets are new $^{40}Ar/^{39}Ar$ ages. Asterisk denotes plateau ages from incremental heating technique, other ages are bulk laser fusion dates (weighted mean of two replicate analyses). Errors are two standard deviations.

N_c, N_m = number of cores drilled, number of samples thermally demagnetizated; N = number of samples contributing to site mean direction; for sites for which no mean direction is available, number of samples with PCA picks is given; D, I = site mean declination, inclination (°); k = dispersion parameter; α_{95} = 95% confidence cone about mean direction (°).

We believe site AZ20 to be the 1.86 Ma flow of Abdel-Monem et al. [16]. The Ribeira Despe-te que Suas (AZ23–AZ26) and the Ribeira do Guilherme (AZ27–AZ28) are both the upper basalts of Fernandez [13], although the Ribeira do Guilherme section is included in an alkali olivine basalt section by Moore [15]. The

relative ages of these two sections were unclear from field relations. Previous mapping and radiometric age determinations led us to expect we had sampled 2–3 Myr of PSV from the Nordeste complex.

4. Paleomagnetic results

Natural remanent magnetization (NRM) directions were measured for all specimens. Stepwise thermal demagnetization was performed on at least 5, and typically 8–11, samples from each site. Temperature steps used were 100º over the interval 0º–200º, 50º over the interval 200º–500º, and 25º above 500º until the sample intensity fell below 1% of the NRM value or the directions became random. The same steps were used for all specimens to enable easy comparison of blocking temperature spectra. At each step susceptibility was measured to monitor progressive changes in magnetic mineralogy during heating. Reverse polarity data show little to moderate normal polarity overprinting, most normal polarity data show virtually no overprinting (Fig. 2).

In this paper we present only the summary results of our thermal demagnetizations, as we consider these data to provide the most robust estimates of site mean directions (many of our samples were too strong to be fully demagnetized using AF demagnetization). Details of the demagnetizations and rock magnetic studies, along with a discussion of the implications of our data for time-averaged field and paleosecular variation models are presented elsewhere (Johnson et al., in prep.; [20]). Demagnetization data were plotted on a vector endpoint diagram [19] — the stable remanence direction is the well-defined component trending to the origin. Strict quantitative criteria were imposed for acceptable stable remanence directions using principal component analysis [21] and Fisher statistics [22]. We require (1) at least five points contributing to the stable remanence direction, (2) the maximum angular deviation (MAD) of the data about the stable component to be less than 5º, (3) within the MAD, the origin must lie on the line defining the stable component in the vector endpoint diagram. Principal components were picked successfully for 90% of the thermally demagnetized samples; only three sites (AZ26, AZ33, AZ38) had more than three thermally demagnetized samples without a clearly defined stable remanence component. Sites AZ01–AZ08 have fewer principal component picks as many of these cores were rather short, yielding only one specimen per core. Fewer samples were thermally demagnetized at these sites as some samples were needed for other experiments (e.g., AF demagnetization, pilot paleointensity experiments, rock magnetic studies).

Site mean directions were calculated for flows with at least five samples with stable remanence directions meeting the criteria above (Table 1, Fig. 3a). These criteria are conservative, but reduce the likeli-

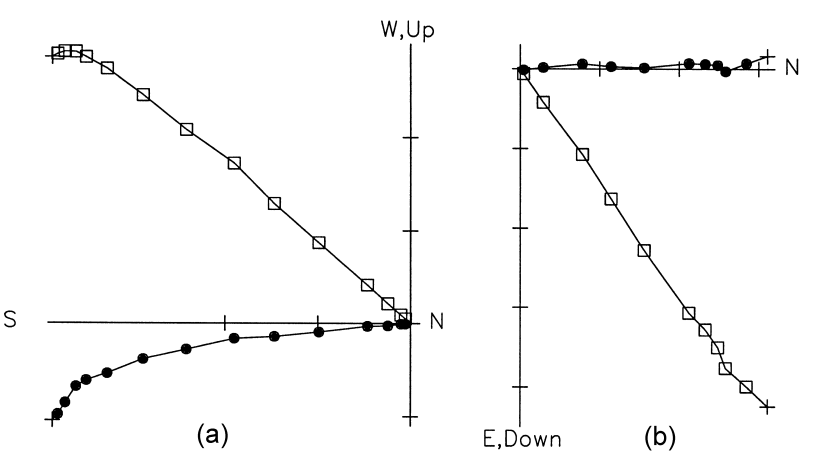


Fig. 2. Typical thermal demagnetization data for (a) reverse and (b) normal polarity sites. Vector endpoint diagram [19]; dots are projections onto horizontal plane, squares are projections onto vertical (north–south) plane. Ticks are 10^{-5} A m^2. Plus signs are NRM directions, temperature steps shown are 100º, then 50º increments from 200º to 500º, then 25º increments.

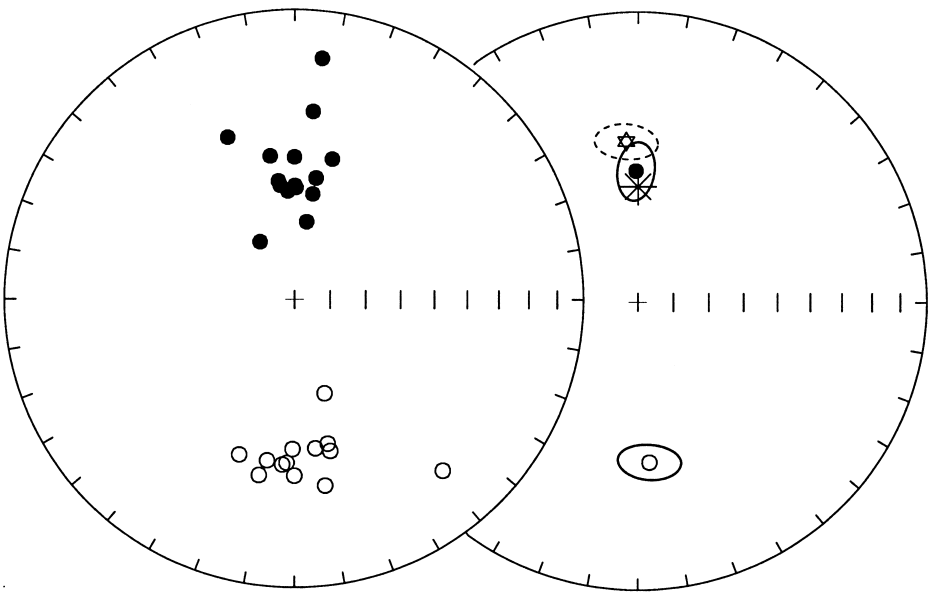


Fig. 3. Site mean paleomagnetic directions computed from at least five (Table 1) fully stepwise thermally demagnetized samples per flow. Stereographic projection, dots are normal polarity site means, circles reverse polarity site means. Figure on the right shows time-averaged normal (dot) and reverse (circle, also antipode as star) polarity field directions. 95% confidence ellipses in the parametric bootstrapped means are plotted rather than α_{95}, since mean directions are non-Fisherian ([23]); normal data: $(\eta_{95}, \zeta_{95}) = (4.9^{o}, 8.7^{o})$; reverse data: $(\eta_{95}, \zeta_{95}) = (5.3^{o}, 8.3^{o})$. Asterisk is geocentric axial dipole (GAD) direction.

hood of estimating inaccurate or biased mean directions. Site mean directions are not available for seven sites: sites AZ02, AZ08 and AZ33 had only four samples (site AZ33 was later supplemented with AF demagnetized data), additionally sites AZ01, AZ03, AZ04 and AZ07 (also AZ02 and AZ08, even when supplemented with AF demagnetized data) had large within-site scatter (k less than 20). Because site mean directions are not available for sites AZ07 and AZ08, the bulk of the normal polarity data span the period 0–100 ka, rather than the intended 0–200 ka.

AZ01–AZ11 and AZ29–AZ38 are normally magnetized, consistent with previous radiometric dates. The normal direction at AZ12 agrees with the early Brunhes Geothermal Project K–Ar dates of 0.76 Ma and 0.78 Ma for two nearby flows (Forjaz, pers. commun.; Geothermal Project K–Ar data). Nordeste sites are predominantly reverse polarity. Our three normal polarity sites from the Nordeste complex are sites AZ15, AZ20 and AZ22.

The TAF directions (vector average of site means) were computed for the normal and reverse polarity data (Table 1, Fig. 3b). The geocentric axial dipole (GAD) prediction for inclination at São Miguel is 57.2°. The normal polarity TAF is not significantly different from GAD (negative inclination anomaly of 4.6°), whilst the reverse TAF is significantly different from a GAD (negative inclination anomaly of 13.5°).

5. $^{40}Ar/^{39}Ar$ ages for the Nordeste complex

We selected twelve reverse polarity sites from the Nordeste complex for $^{40}Ar/^{39}Ar$ dating (Table 1), from three separate cliff sections along the coastline. Younger samples were not analyzed because most have already been dated [17,18]. Thin sections were examined for the effects of hydrothermal alteration, glass and the presence of mafic phenocrysts such as olivine and pyroxene. Our analyses were carried out at the Vrije University of Amsterdam and included bulk fusion and incremental heating experiments [24,25]. Eight sites (Table 2) were dated using the bulk fusion technique on 5-mm diameter, 1-mm-thick cores. These samples contained intermediate to coarse plagioclase. The remaining four sites

Table 2
Summary of laser fusion experiments

Sample (file)	^{36}Ar	^{37}Ar	^{38}Ar	^{39}Ar	^{40}Ar	$\%^{40}Ar^{*}$	$^{40}Ar^{*}/^{39}Ar(K)$	J	Age	SD	Mean age	SD
AZ14 (025)	0.00861	5.1659	−0.0077	5.9983	8.9400	71.5	1.0635	0.000447	0.857	0.012	0.846	0.011
AZ14 (026)	0.00930	5.7412	−0.0025	6.8292	9.8391	72.0	1.0385	0.000447	0.837	0.010		
AZ16T (027)	0.04834	6.9443	−0.0002	0.8999	14.9964	81.0	1.0969	0.000423	0.838	0.007	0.843	0.009
AZ16T (028)	0.00258	4.8958	−0.0030	4.9753	6.3412	88.0	1.1212	0.000423	0.856	0.010		
AZ18J (029)	0.00264	4.4082	−0.0019	4.5753	6.2020	87.4	1.1850	0.000398	0.850	0.009	0.849	0.014
AZ18J (030)	0.00183	2.4347	−0.0018	2.1556	3.0581	46.2	1.1682	0.000398	0.838	0.023		
AZ19M (031)	0.00278	4.2982	−0.0031	4.4963	6.5439	87.4	1.2727	0.000372	0.853	0.009	0.852	0.010
AZ19M (032)	0.00347	5.3120	−0.0027	5.0824	7.5155	86.3	1.2770	0.000369	0.851	0.008		
AZ21 (033)	0.03871	4.3370	0.0010	2.3049	14.6754	22.1	1.4040	0.000344	0.871	0.013	0.878	0.045
AZ21 (034)	0.03985	4.4994	0.0017	2.2345	15.0033	21.5	1.4451	0.000341	0.890	0.016		
AZ24I (035)	0.00612	4.9461	0.0008	5.4158	7.6712	76.4	1.0827	0.000425	0.829	0.013	0.817	0.014
AZ24I (036)	0.00516	4.9206	0.0010	5.5889	7.4585	79.5	1.0618	0.000423	0.809	0.011		
AZ25 (037)	0.00297	4.1098	0.0006	4.0607	5.4890	84.0	1.1359	0.000401	0.821	0.012	0.814	0.017
AZ25 (038)	0.01376	6.7719	−0.0007	6.9249	12.1369	66.5	1.1655	0.000380	0.799	0.015		
AZ26A (039)	0.00639	5.5309	0.0026	7.5995	10.8412	34.1	1.1781	0.000376	0.799	0.008	0.793	0.008
AZ26A (040)	0.01008	6.3865	0.0016	9.4424	14.0212	36.8	1.1695	0.000374	0.789	0.007		

Table 3
Incremental heating results

Site	Total fusion age (ka)	Plateau age (ka)	(% Gas)	Inverse isochron age (ka)	$^{40}Ar/^{36}Ar$ intercept
AZ13	842 ± 24	835 ± 11	(95.6)	831 ± 5	296.4 ± 3.2
AZ17	820 ± 28	819 ± 17	(100)	820 ± 7	294.4 ± 6.8
AZ23	790 ± 16	785 ± 11	(97.2)	787 ± 5	292.3 ± 4.6
AZ28	790 ± 24	776 ± 12	(86.5)	781 ± 5	289.9 ± 3.5

Isochrons were calculated over the steps which contributed to the plateau age. Uncertainties are two standard errors. Detailed results in Appendices A, B, C and D. Correction factors used were as follows: $^{40/39}K = 0.00086$; $^{36/37}Ca = 0.00026$; $^{39/37}Ca = 0.00067$; $^{40/36}$(atmospheric Ar) = 295.5; K/Ca = 0.49; $J = 0.000328 \pm 0.000001$.

(Table 3) were dated using the stepwise laser incremental heating technique on approximately 30-mg groundmass separates (e.g., [26]), an appropriate technique because these samples contain significant amounts of olivine and pyroxene phenocrysts, minerals known to have very low K concentrations and possibly excess argon.

The samples were irradiated for 1 h in the CLICIT facility at the Oregon State University TRIGA reactor along with Taylor Creek Rhyolite sanidine (standard with an age of 27.92 Ma, cross-calibrated to 162.9 Ma SB-3 biotite [27]). In the incremental heating experiments the laser power was increased from 1 W to 8 W typically over 7–8 steps, with the laser beam defocussed to approximately a 3-mm spot. The 30-mg groundmass separates were spread thinly on the bottom of 13-mm diameter copper pans, and homogeneous heating was obtained by slowly circling the laser beam over the bottom of the pan. Corrections for interfering reactions from K and Ca were those reported by Wijbrans et al. [25], radioactive decay of ^{37}Ar and ^{39}Ar [24], and mass spectrometer discrimination (1.003 ± 0.1% pmu) determined from repeated analysis of small aliquots of air argon. Blank corrections were estimated for each experiment individually by interpolation between values measured between every two or three unknowns. The calculation of plateau ages required data from at least three temperature steps, which together comprise more than 50% of the total argon release, with

each contributing step yielding an age within two standard deviations of the mean age (Tables 1 and 2 and Appendices A, B, C and D). Bulk fusion experiments include three heating steps on duplicate chips from the same sample, but because both the first step and the final fusion step contain large proportions of air argon they were omitted from the age calculation. Most bulk fusion age determinations are obtained from cores with very high percentages of radiogenic argon (typically greater than 75% radiogenic argon, Table 3); only two sites (AZ21, AZ26) have less than 50% radiogenic argon in both samples analyzed. In addition, results from the groundmass incremental heating experiments demonstrate consistently good plateaus for the majority of the intermediate temperature steps, providing reassurance that the bulk fusion ages from the single intermediate temperature step are reliable.

$^{40}Ar/^{39}Ar$ age determinations were obtained for all 12 sites (Tables 1–3). The oldest mean age of 0.878 Ma for site AZ21 is from a sample with lower radiogenic ^{40}Ar than typically obtained (Table 3) and is thus less reliable than other results. However, the age is consistent with our other $^{40}Ar/^{39}Ar$ results and with the regional geology: the site is stratigraphically low, at the base of the Porto de Nordeste cliff section.

The average age for the Praia Lombo da Gordo section (AZ13–AZ19) is 0.847 Ma. Field stratigraphic relations indicate the age progression up the section (decreasing age) is AZ13, AZ14, AZ18, AZ16, AZ17, AZ19 (Table 1). Bulk fusion ages for AZ14, AZ16, AZ18 and AZ19 do not provide a clear trend, but are identical within the analytical uncertainties. The more precise mean plateau ages and inverse isochron mean ages for AZ13 and AZ17 indicate a younger age for AZ17 relative to AZ13, consistent with field stratigraphic relations, though again the age determinations are indistinguishable within the uncertainties. $^{40}Ar/^{36}Ar$ intercept values are indistinguishable from that of air (295.5) indicating no extraneous argon is present.

AZ23–AZ26 are from the Ribeira Despe-te que Suas section and give mean ages of 0.79–0.82 Ma. The relative ages of AZ24, AZ25 and AZ26 are consistent with stratigraphic relations within the Ribeira Despe-te que Suas section. The younger age of AZ23 with respect to AZ24 is not consistent with field observations; however, the sense and magnitude of the age offset between the incremental heating age determination for AZ23 and the bulk fusion age determination for AZ24 is consistent with the small bias observed for the Praia Lombo da Gordo sites between the more precise incremental heating ages and the bulk fusion dates.

Age estimates for site AZ28 are 0.78–0.79 Ma, very similar to the incremental heating age determinations for site AZ23. This is consistent with our field interpretation that both the Ribeira Despe-te que Suas and the Ribeira do Guilherme sections are part of the upper basalt sequence, and should have younger ages on average than the lower basalts of the Porto de Nordeste and Praia Lombo da Gordo sections.

6. Discussion and conclusions

Our paleomagnetic data from the western part of São Miguel are consistent with previous Brunhes age radiometric dates for this part of the island. With the exception of site AZ12, which, as mentioned earlier is likely to be of early Brunhes age, our western São Miguel sites span the period 0–0.1 Ma.

Reverse polarity field directions were obtained from all flows sampled in the Nordeste complex, except sites AZ15, AZ20 and AZ22. Previous K–Ar analyses suggested an age range of 4–0.95 Ma for the Nordeste complex; however, our reverse polarity paleomagnetic data and $^{40}Ar/^{39}Ar$ ages are consistent with a late Matuyama age for most of the exposed eastern part of São Miguel. Site AZ21 is the lowermost flow in the Porto de Nordeste cliff section, and is part of the lower basalt sequence of the Nordeste complex [13]. Abdel-Monem et al. [16] inferred an age of 4 Ma for this unit in contrast to our $^{40}Ar/^{39}Ar$ result of 0.878 Ma. Abdel-Monem et al. [16] obtained a second K–Ar age for the lower basalts of 3.17 Ma. Our field observations indicate that the Praia Lombo da Gordo section is part of the lower basalts (in agreement with Fernandez [13]); we obtain $^{40}Ar/^{39}Ar$ ages of 0.820–0.852 Ma for this section, similar to the 0.90 Ma and 0.92 Ma K–Ar dates obtained from two cliff section flows to the south at Agua Retorta (Forjaz, pers. commun.; Geothermal Project data). The age of the upper basalt sequence was previously suggested to be 1.86

Ma [16] based on K–Ar dating at our site AZ20. We did not obtain radiometric dates from this site; however, we did obtain $^{40}Ar/^{39}Ar$ ages of 0.79–0.82 Ma for an upper basalt section at Ribeira Despe-te que Suas (sites AZ23–AZ26).

We sampled two normal polarity flows (AZ20, AZ22) and one normal polarity dike (AZ15) in the Nordeste complex. AZ20 is an ankaramite flow, dated by Abdel-Monem et al. [16] at 1.86 Ma. AZ22 is an ankaramite, intra-canyon flow, identified in the field as likely to be much younger than the other flows in the Porto de Nordeste section. Both AZ20 and AZ22 are probably part of the Nordeste ankaramites, which unconformably overly the lower basalts and gradually transit into the upper basalts [13]. On the basis of our new $^{40}Ar/^{39}Ar$ ages which indicate that the lower basalts of the Porto de Nordeste and Praia Lombo da Gordo sections are late Matuyama, it is quite possible that the previous age determination for AZ20 is incorrect and that AZ20 and AZ22 are Brunhes age flows. We have no radiometric age determinations for site AZ15, though it is part of a normal polarity dike swarm intruding the late Matuyama age Praia Lombo da Gordo section, and could be of Brunhes age. Age determinations for the reverse polarity upper basalt flows sampled from the Ribeira Despe-te que Suas and the Ribeira do Guilherme sections, indicate that these sections were emplaced close to the Brunhes/Matuyama boundary.

Our age determinations indicate that most of the exposed Nordeste complex was emplaced over a period of 0.1 Myr, as opposed to the previously suggested 2.15 Myr, and that there may have been considerable temporal overlap in the five stratigraphic sequences of Fernandez [13] comprising the Nordeste complex. In particular our results indicate that in some places the Nordeste ankaramites may be younger than parts of the upper basalt sequence. Radiometric dates for our sites AZ20 and AZ22 are necessary to confirm this latter interpretation.

Many recent studies based on modeling and careful examination of existing data sets (most recently [1,3,8,28–30]; review in Merrill et al. [31]) have resurrected long-held concerns that most currently available paleomagnetic data from 0 to 5 Ma lava flows are inadequate for detailed studies of PSV and the TAF. Data quality issues arise because most existing PSVL data were collected in the 1970s; demagnetization techniques used to analyze these data were less refined, and principal component analysis had not been adopted. However, inferences of non-GAD field structure (e.g., Fig. 3b) and the significance of normal/reverse polarity asymmetries clearly require very accurate estimates of paleomagnetic field directions.

Temporal sampling problems arise because commonly paleomagnetic studies on 0–5 Ma rocks included at most the measurement of one or two K–Ar ages. Based on previous K–Ar results we had hoped to sample approximately 2–3 Myr of secular variation from the Nordeste complex, but more numerous and more accurate $^{40}Ar/^{39}Ar$ ages demonstrate significant biases in the existing K–Ar age determinations toward older ages and a much longer time interval for emplacement of the complex. K–Ar ages can be erroneously old or young, depending on whether alteration and/or argon diffusion out of the sample has occurred. The earlier K–Ar dates for São Miguel [16] were calculated from samples with low percentages of radiogenic argon (less than 16% radiogenic argon, compared with typical values here of 65%–90%, Table 3), and are less reliable than our measurements. Other studies which have compared $^{40}Ar/^{39}Ar$ and earlier K–Ar data for sequences of lava flows, have found a similar decrease in the spread of ages obtained using $^{40}Ar/^{39}Ar$. A study of the Karoo basalts in Africa found $^{40}Ar/^{39}Ar$ ages tightly clustered around 180 Ma, in contrast to the previous K–Ar determinations spanning 155–210 Ma [32]. Instead of sampling the period 1–4 Ma, our Nordeste paleomagnetic data provide information on geomagnetic field behavior over the period 0.78–0.88 Ma.

Thus our geomagnetic polarity results and $^{40}Ar/^{39}Ar$ ages from São Miguel have important implications for the timing and duration of formation of the Nordeste complex, regional melt generation processes, and temporal sampling of the geomagnetic field. The eastern part of São Miguel is only slightly older than the volcanically active western portion of the island, and suggests that communication between magma source regions in the western and eastern portions of São Miguel could contribute to the observed isotopic gradients across the island [10,11].

Paleomagnetic data and radiometric dates allow the calculation of a reverse polarity TAF direction

representing the period 0.78–0.88 Ma and a normal polarity TAF direction likely representing the period 0–0.78 Ma. A significant fraction (11 of our 15 flows) of our normal polarity sites span the period 0–0.1 Ma: the directions for this smaller subset of data have a Fisherian distribution and calculation of the corresponding TAF direction yields a mean declination and inclination of respectively 0.9º and 49.0º, with a k of 30 and an α_{95} of 8.5º. As for the complete normal polarity data set the 0–0.1 Ma subset has a mean direction not significantly different from either a GAD or the reverse polarity TAF direction.

Our results show that the normal polarity TAF direction at São Miguel over either 0–0.1 Ma or 0–0.78 Ma, and the reverse polarity TAF direction for 0.78–0.88 Ma are not significantly different. Deviations from a GAD field are larger for the reverse polarity TAF than the normal polarity TAF. It is clear that high-precision $^{40}Ar/^{39}Ar$ age determinations from a substantial fraction of flows sampled in a given paleomagnetic study are essential for accurate inferences of long-term geomagnetic field behavior.

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Appendix A. Incremental heating data for AZ13

			$^{36}Ar(a)$	^{37}Ar	^{38}Ar	$^{39}Ar(K)$	$^{40}Ar(a+r)$	Age $\pm 2\sigma$	$^{40}Ar(r)$ (%)	Fraction (%)	K/Ca
96M0340B	1.00 W	0	0.001420	0.000000	0.000086	0.016198	0.450390	1.37 ± 2.10	6.86	0.11	No calcium
96M0340C	2.00 W	0	0.001741	0.171121	0.000102	0.138217	0.700629	0.97 ± 0.16	26.55	0.96	0.396
96M0340E	3.00 W	0	0.001426	0.518919	−0.000010	0.478884	0.989295	0.85 ± 0.04	57.37	3.32	0.452
96M0340F	4.00 W	1	0.001137	1.035305	−0.000117	1.120825	1.921289	1.02 ± 0.03	82.48	7.77	0.530
96M0340H	5.00 W	1	0.001489	2.302153	−0.000636	3.275578	4.273463	0.84 ± 0.00	89.64	22.72	0.697
96M0340I	6.00 W	1	0.001963	2.840310	−0.001038	4.619286	5.780808	0.81 ± 0.01	89.91	32.04	0.797
96M0340K	7.00 W	1	0.004705	5.111556	0.000364	2.971224	4.761618	0.81 ± 0.02	70.76	20.61	0.285
96M0340L	8.00 W	1	0.051094	6.069273	0.000252	1.317490	16.661750	0.85 ± 0.09	9.38	9.14	0.106
96M0340E	3.00 W	1	0.001391	0.531117	−0.000000	0.478667	0.987629	0.86 ± 0.04	58.37	3.32	0.442
Total			0.066365	18.579755	−0.000998	14.416369	36.526871				

Summary results

	Average K/Ca	Cumulative fraction	Total	MWSD	$^{40}Ar(a+r)/^{36}Ar(a) \pm 2\sigma$	$^{40}Ar^{*}/^{39}Ar(K) \pm 2\sigma$	Age	2σ
Age plateau	0.46	95.61%				1.1630 ± 0.0146	0.835	0.011
Total fusion						1.1734 ± 0.0173	0.842	0.012
Normal isochron			39.78	13.26	296.4 ± 3.2	1.1538 ± 0.0069	0.828	0.005
Inverse isochron			43.51	14.50	296.4 ± 3.2	1.1579 ± 0.0064	0.831	0.005

Note. Top table gives data as a function of heating step. In the third column 1/0 indicates whether step included/excluded from plateau age and isochron age calculation. Summary part gives plateau, total fusion and normal ($^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$) and reverse ($^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$) isochron ages.

Appendix B. Incremental heating results for AZ17

			^{36}Ar(a)	^{37}Ar	^{38}Ar	^{39}Ar(K)	^{40}Ar(a + r)	Age ± 2σ	^{40}Ar(r) (%)	Fraction (%)	K/Ca
96M0341B	1.00 W	1	0.001585	0.000000	−0.000140	0.027926	0.494539	0.657 ± 1.204	5.29	0.33	No calcium
96M0341C	2.00 W	1	0.001473	0.492048	0.000058	0.290961	0.751557	0.763 ± 0.060	42.09	3.44	0.290
96M0341E	3.00 W	1	0.000724	0.885585	−0.000203	0.633802	0.956111	0.822 ± 0.044	77.59	7.49	0.351
96M0341F	4.00 W	1	0.000671	1.044933	−0.000355	0.977921	1.340127	0.819 ± 0.025	85.14	11.55	0.459
96M0341H	5.00 W	1	0.002192	3.001138	−0.001154	3.540937	4.704078	0.804 ± 0.011	86.17	41.83	0.578
96M0341I	6.00 W	1	0.002995	3.949423	0.000200	2.143450	3.275573	0.783 ± 0.016	72.94	25.32	0.266
96M0341K	7.00 W	1	0.002465	2.952316	0.000460	0.445043	1.223898	0.781 ± 0.078	40.47	5.26	0.074
96M0341L	8.00 W	1	0.008772	7.052588	0.000741	0.404384	3.091853	0.867 ± 0.120	16.16	4.78	0.028
Total			0.020877	19.378031	−0.000393	8.464424	15.837737				

Summary results

	Average K/Ca	Cumulative fraction	Total	MWSD	^{40}Ar(a + r)/^{36}Ar(a) ± 2σ	^{40}Ar*/^{39}Ar(K) ± 2σ	Age	2σ
Age plateau	0.29	100.00%				1.1410 ± 0.0239	0.819	0.017
Total fusion						1.1423 ± 0.0200	0.820	0.014
Normal isochron			13.59	2.26	294.4 ± 6.8	1.1394 ± 0.0110	0.818	0.008
Inverse isochron			14.51	2.42	294.8 ± 6.8	1.1417 ± 0.0104	0.820	0.007

See note Appendix A.

Appendix C. Incremental heating results for AZ23

			^{36}Ar(a)	^{37}Ar	^{38}Ar	^{39}Ar(K)	^{40}Ar(a + r)	Age ± 2σ	^{40}Ar(r) (%)	Fraction (%)	K/Ca
96M0342B	1.00 W	0	0.003488	0.000000	−0.000126	0.415353	1.726707	1.043 ± 0.053	40.30	2.78	No calcium
96M0342C	2.00 W	1	0.003445	2.864748	−0.000989	4.043900	6.207279	0.798 ± 0.009	83.55	27.03	0.692
96M0342E	3.00 W	1	0.002476	3.414844	−0.002876	5.879031	8.176422	0.788 ± 0.008	91.00	39.30	0.844
96M0342F	4.00 W	1	0.001083	1.245619	−0.000551	1.931480	2.718998	0.773 ± 0.013	88.18	12.91	0.760
96M0342H	5.00 W	1	0.000390	0.480385	−0.000313	0.542753	0.804639	0.791 ± 0.040	85.64	3.63	0.554
96M0342I	6.00 W	1	0.003113	3.301985	0.000452	1.676218	2.935576	0.748 ± 0.014	68.63	11.20	0.249
96M0342L	7.00 W	1	0.023262	6.409247	0.001649	0.471816	7.444850	0.753 ± 0.153	7.67	3.15	0.036
Total			0.037257	17.716828	−0.002754	14.960551	30.014469				

Summary results

	Average K/Ca	Cumulative fraction	Total	MWSD	^{40}Ar(a + r)/^{36}Ar(a) ± 2σ	^{40}Ar*/^{39}Ar(K) ± 2σ	Age	2σ
Age plateau	0.52	97.22%				1.2618 ± 0.0170	0.785	0.011
Total fusion						1.2703 ± 0.0122	0.790	0.008
Normal isochron			51.84	12.96	291.5 ± 4.6	1.2595 ± 0.0088	0.784	0.005
Inverse isochron			50.38	12.59	292.3 ± 4.6	1.2644 ± 0.0078	0.787	0.005

See note Appendix A.

Appendix D. Incremental heating results for AZ28

			^{36}Ar(a)	^{37}Ar	^{38}Ar	^{39}Ar(K)	^{40}Ar(a + r)	Age ± 2σ	^{40}Ar(r) (%)	Fraction (%)	K/Ca
96M0344B	1.00 W	0	0.006304	0.000000	−0.000112	0.019804	1.931941	2.063 ± 1.894	3.58	0.17	No calcium
96M0344C	2.00 W	0	0.013421	0.750620	0.000419	0.671906	4.961192	0.876 ± 0.069	20.06	5.61	0.439
96M0344D	3.00 W	1	0.003126	1.353303	−0.001000	2.033893	3.560645	0.767 ± 0.016	74.02	16.99	0.736
96M0344G	4.00 W	1	0.001854	0.606017	−0.000348	1.255036	2.187056	0.773 ± 0.016	74.91	10.49	1.015
96M0344H	5.00 W	1	0.001673	0.822001	−0.000273	2.075360	3.233365	0.781 ± 0.007	84.66	17.34	1.237
96M0344I	6.00 W	1	0.005358	1.149019	−0.000150	2.679123	5.009620	0.757 ± 0.015	68.36	22.38	1.143
96M0344L	7.00 W	1	0.021458	3.974449	0.000917	2.303719	9.277972	0.754 ± 0.019	31.65	19.25	0.284
96M0344M	8.00 W	0	0.057861	5.466185	0.001449	0.930736	18.629536	0.974 ± 0.080	8.22	7.78	0.083
Total			0.111056	14.121594	0.000901	11.969576	48.791327				

Summary results

	Average K/Ca	Cumulative fraction	Total	MWSD	^{40}Ar(a + r)/^{36}Ar(a) ± 2σ	^{40}Ar*/^{39}Ar(K) ± 2σ	Age	2σ
Age plateau	0.71	86.45%				1.3114 ± 0.0154	0.776	0.009
Total fusion						1.3346 ± 0.0196	0.790	0.012
Normal isochron			10.15	3.38	289.9 ± 3.5	1.3187 ± 0.0086	0.780	0.005
Inverse isochron			9.57	3.19	289.9 ± 3.5	1.3195 ± 0.0084	0.781	0.005

See note Appendix A.

References

[1] M.W. McElhinny, P.L. McFadden, Paleosecular variation over the past 5 Ma, Geophys. J. Int. (1997) (submitted).

[2] X. Quidelleur, J.-P. Valet, V. Courtillot, G. Hulot, Long-term geometry of the geomagnetic field for the last 5 million years; an updated secular variation database from volcanic sequences, Geophys. Res. Lett. 21 (1994) 1639–1642.

[3] C.L. Johnson, C.G. Constable, Palaeosecular variation recorded by lava flows over the last 5 million years, Philos. Trans. R. Soc. London 354 (1996) 89–141.

[4] H. Saucier, A. Roche, Etude paleomagnetique de laves de Faial et de Graciosa, Commun. Geol. Serv. Port. 48 (1964) 5–13.

[5] J. Serughetti, A. Roche, Etude paleomagnetique de laves de l'île de Flores, C. R. Acad. Sci., Paris 267 (1968) 1185–1188.

[6] M.A. Salgueiro, Paleomagnetic study and chronological stratigraphy of volcanic formations from Santa Maria Island, Azores , Arquipélago, Life Earth Sci. 9 (1991) 83–99.

[7] B.B. Ellwood, N.D. Watkins, C. Amerigian, S. Self, Brunhes Epoch Geomagnetic Secular Variation on Terceira Island, Central North Atlantic, J. Geophys. Res. 78 (1973) 8699–8710.

[8] C.L. Johnson, C.G. Constable, The time-averaged geomagnetic field as recorded by lava flows over the last 5 Ma, Geophys. J. Int. 122 (1995) 489–519.

[9] J.-G. Schilling, Azores mantle blob: rare-earth evidence, Earth Planet. Sci. Lett. 25 (1975) 103–115.

[10] C.J. Hawkesworth, N.W. Rogers, P.W.C. van Calstren, M.A. Menzies, ^{143}Nd/^{144}Nd and ^{87}Sr/^{86}Sr ratios from the Azores and their significance in LIL-element enriched mantle, Nature 280 (1979) 28–31.

[11] E. Widom, R.W. Carlson, J.B. Gill, H.-U. Schmincke, Th–Sr–Nd–Pb isotope and trace element evidence for the origin of São Miguel, Azores enriched mantle source, Chem. Geol. 140 (1997) 49.

[12] B. Booth, R. Croasdale, G.P.L. Walker, A quantitative study of five thousand years of volcanism on São Miguel, Azores, Philos. Trans. R. Soc. London 288 (1978) 271–319.

[13] L.A. Fernandez, Geology and petrology of the Nordeste volcanic complex, São Miguel, Azores: summary, Geol. Soc. Am. Bull. 91 (1980) 675–680.

[14] R.B. Moore, Volcanic geology and eruption frequency, São Miguel, Azores, Bull. Volcanol. 52 (1990) 602–614.

[15] R.B. Moore, Geology of three late Quaternary stratovolcanoes on São Miguel, Azores, U.S.G.S. Bull., 1900.

[16] A.A. Abdel-Monem, L.A. Fernandez, G.M. Boone, K–Ar ages from the eastern Azores group (Santa Maria, São Miguel and the Formigas Islands), Lithos 8 (4) (1975) 247–254.

[17] E.H. McKee, R.B. Moore, Potassium–argon dates for trachytic rocks on São Miguel, Azores, Isochron/West 58 (1992) 9–11.

[18] R.B. Moore, M. Rubin, Radiocarbon dates for lava flows and pyroclastic deposits on São Miguel, Azores, Radiocarbon 33 (1991) 151–164.

[19] J.D. Zijderveld, A.C. demagnetization in rocks: analysis of results, in: Collinson, Creer, Runcorn (Eds.), Methods in Paleomagnetism, Elsevier, Amsterdam, 1967, pp. 254–286.
[20] C.L. Johnson, C.G. Constable, J. Gee, H. Staudigel, L. Tauxe, M. Salgueiro, V.H. Forjaz, Paleomagnetism of 0–4 Ma lavas from São Miguel, Azores, Eos 77 (1996) 165.
[21] J.L. Kirschvink, The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[22] R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A 217 (1953) 295–305.
[23] L. Tauxe, Paleomagnetic Principles and Practice, Kluwer, Dordrecht, 1998 (in press).
[24] I.M. McDougall, T.M. Harrison, Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method, Oxford University Press, Oxford, 1988, 212 pp.
[25] J.R. Wijbrans, M.S. Pringle, A.A. Koppers, R. Scheveers, Argon geochronology of small samples using the VULKAAN argon laser probe, Proc. K. Akad. Wet. 98 (1995) 185–218.
[26] A.A.P. Koppers, H. Staudigel, J.R. Wijbrans, $^{40}Ar/^{39}Ar$ ages and isotope geochemistry of Vlinder Guyot and adjacent seamounts (W. Pacific Plate): from altered submarine basalts to mantle source characteristics, J. Conf. Abstr. 1 (1996) 325.
[27] W.A. Duffield, G.B. Dalrymple, The Taylor Creek Rhyolite of New Mexico: a rapidly emplaced field of lava domes and flows, Bull. Volcanol. 52 (1990) 475–487.
[28] J. Carlut, V. Courtillot, How complex is the time-averaged geomagnetic field over the last 5 million years? Geophys. J. Int. (1997) (submitted).
[29] C.L. Johnson, C.G. Constable, The time-averaged geomagnetic field: global and regional biases for 0–5 Ma, Geophys. J. Int. 131 (1997) 643–666.
[30] M.W. McElhinny, P.L. McFadden, R.T. Merrill, The time-averaged geomagnetic field 0–5 Ma, J. Geophys. Res. 101 (1996) 25007–25027.
[31] R.T. Merrill, M.W. McElhinny, P.L. McFadden, The Magnetic field of the Earth: Paleomagnetism, the Core and the Deep Mantle, Academic Press, San Diego, CA, 1996.
[32] J.S. Marsh, P.R. Hooper, J. Rehacek, R.A. Duncan, A.R. Duncan, Stratigraphy and age of Karoo basalts of Lesotho and implications for correlations within the Karoo igneous province, in: Large Igneous Provinces: Continental, Oceanic and Planetary Flood Volcanism, Am. Geophys. Union, Geophys. Monogr. 100 (1997), 247 pp.